1. Write a program to demonstrate the basic commands of Windows/Linux OS, display the directory structure, and monitor running processes using process monitoring tools. CO1 (K2)

Aim & Algorithm (20)	Coding (30)	Output & Result (30)	Viva-Voce (10)	Record (10)	Total (100)

Signature of the Examiner

2. Write a C/Java program to implement and simulate the CPU scheduling algorithms FCFS, SJF (Non-Preemptive), Priority Scheduling, and Round Robin, and calculate the average waiting time and turnaround time. CO2 (K4)

Aim & Algorithm (20)	Coding (30)	Output & Result (30)	Viva-Voce (10)	Record (10)	Total (100)

Signature of the Examiner

3. Write a C program to simulate process management by performing process creation, termination, and context switching using system calls (fork(), exec(), wait(), exit()). CO2 (K4)

Aim & Algorithm (20)	Coding (30)	Output & Result (30)	Viva-Voce (10)	Record (10)	Total (100)

Signature of the Examiner

4. Write a C program to implement Inter-Process Communication (IPC) using pipes, shared memory, and message queues. CO2 (K4)

Aim & Algorithm (20)	Coding (30)	Output & Result (30)	Viva-Voce (10)	Record (10)	Total (100)

Signature of the Examiner

5. Write a C/Java program to solve the Producer–Consumer problem using Mutex and Semaphores. CO2 (K4)

Aim & Algorithm (20)	Coding (30)	Output & Result (30)	Viva-Voce (10)	Record (10)	Total (100)

Signature of the Examiner

6. Write a C/Java program to implement the Banker's Algorithm for deadlock avoidance and check whether the system is in a safe state. CO2 (K4)

Aim & Algorithm (20)	Coding (30)	Output & Result (30)	Viva-Voce (10)	Record (10)	Total (100)

Signature of the Examiner

7. Write a C/Java program to simulate contiguous memory allocation, paging, and segmentation techniques. CO3 (K3)

Aim & Algorithm (20)	Coding (30)	Output & Result (30)	Viva-Voce (10)	Record (10)	Total (100)

Signature of the Examiner

8. Write a C/Java program to implement the FIFO, LRU, and Optimal page replacement algorithms and find the number of page faults. CO3 (K3)

Aim & Algorithm (20)	Coding (30)	Output & Result (30)	Viva-Voce (10)	Record (10)	Total (100)

Signature of the Examiner

9. Write a Java program to perform file handling operations such as creating, writing, reading, deleting, and closing a file using Java I/O classes, and handle file permissions. CO4 (K2)

Aim & Algorithm (20)	Coding (30)	Output & Result (30)	Viva-Voce (10)	Record (10)	Total (100)

Signature of the Examiner

10. Write a C/Java program to implement the disk scheduling algorithms FCFS, SSTF, SCAN, and C-SCAN, and calculate the total head movement. CO4 (K2)

Aim & Algorithm (20)	Coding (30)	Output & Result (30)	Viva-Voce (10)	Record (10)	Total (100)

Signature of the Examiner